Traffic Congestion Dataset - Columns

Columns

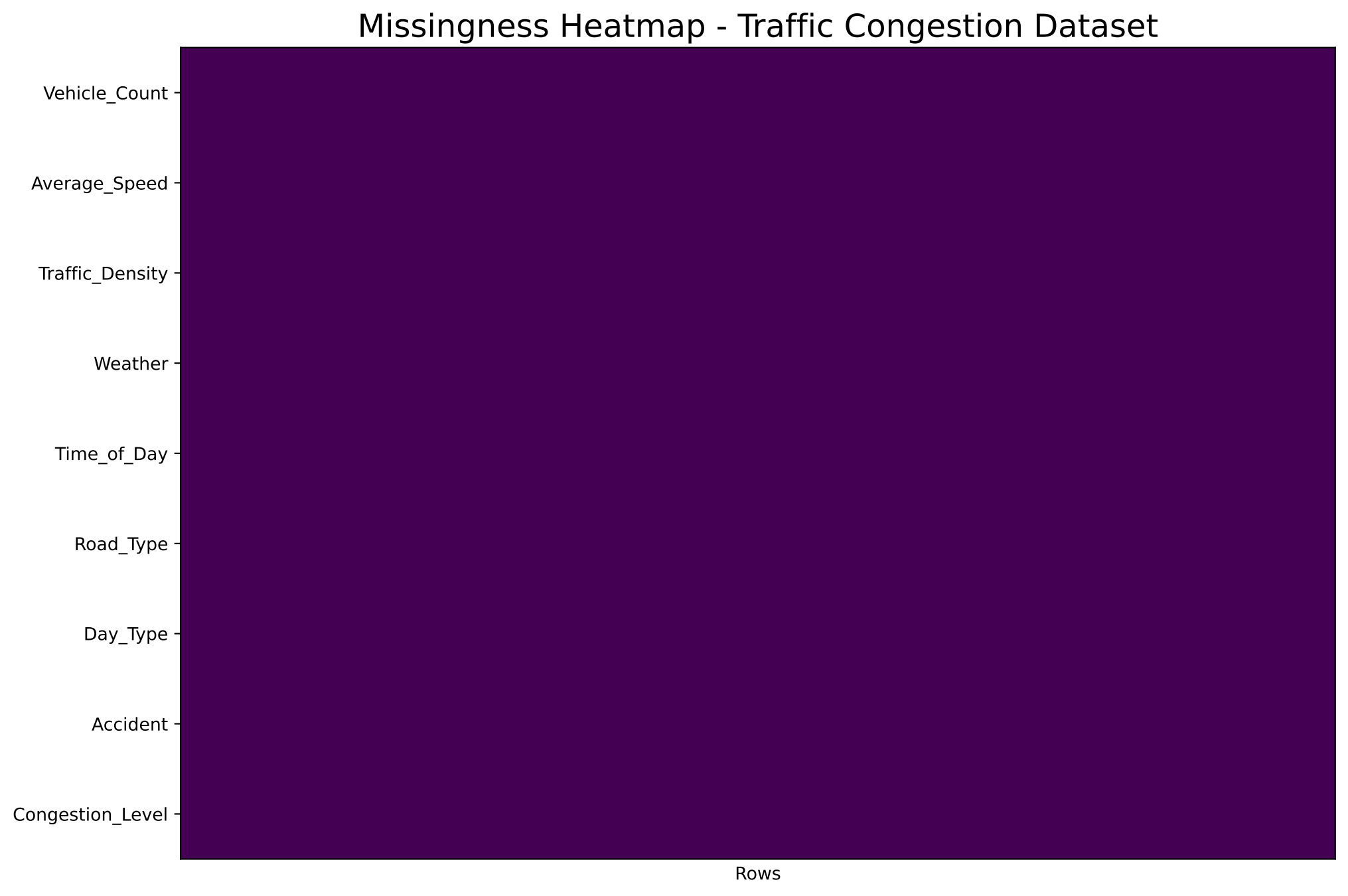
1. Vehicle_Count
2. Average_Speed
3. Traffic_Density
4. Weather
5. Time_of_Day
6. Road_Type
7. Day_Type
8. Accident
9. Congestion_Level

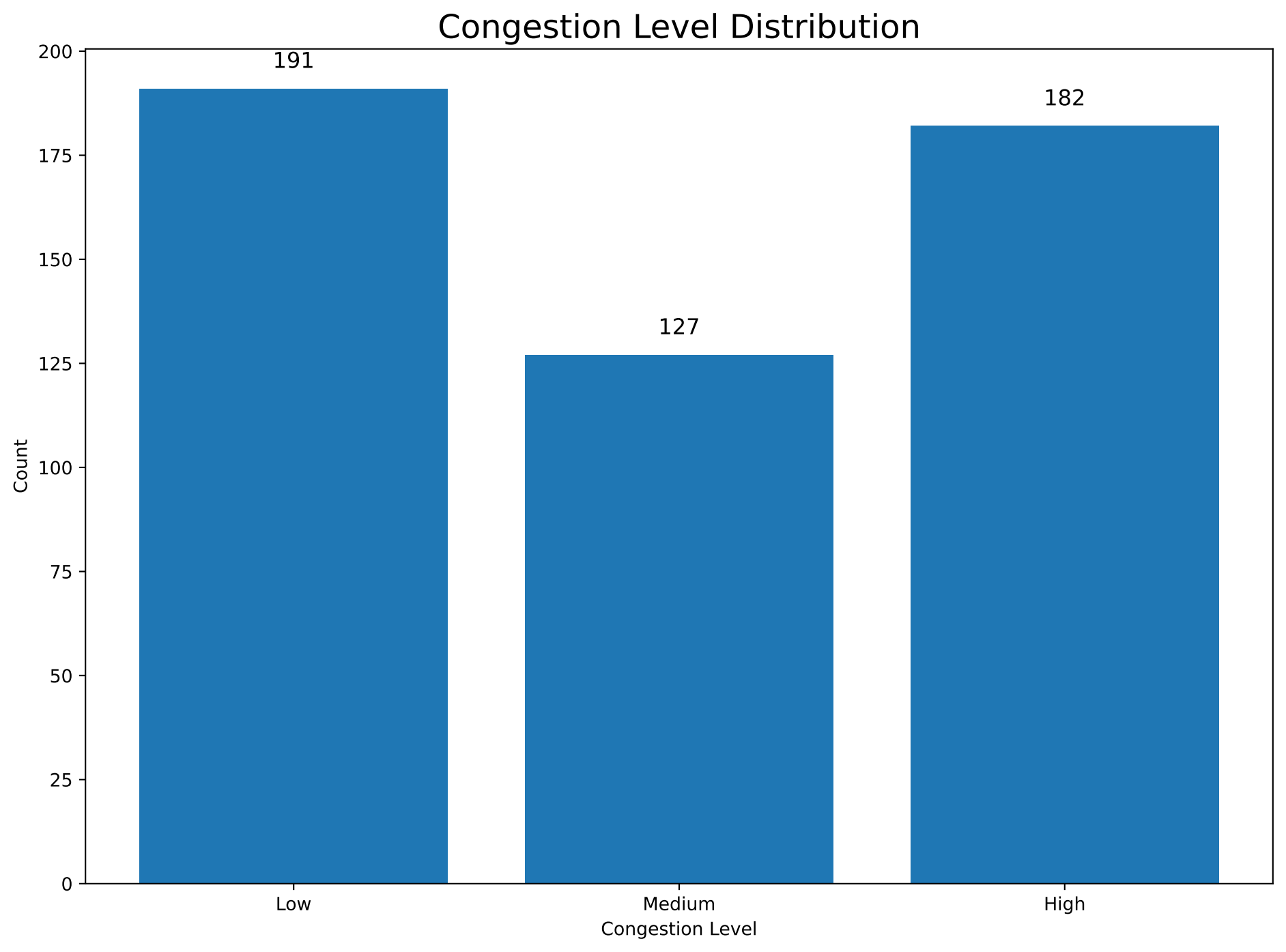
Dataset Size

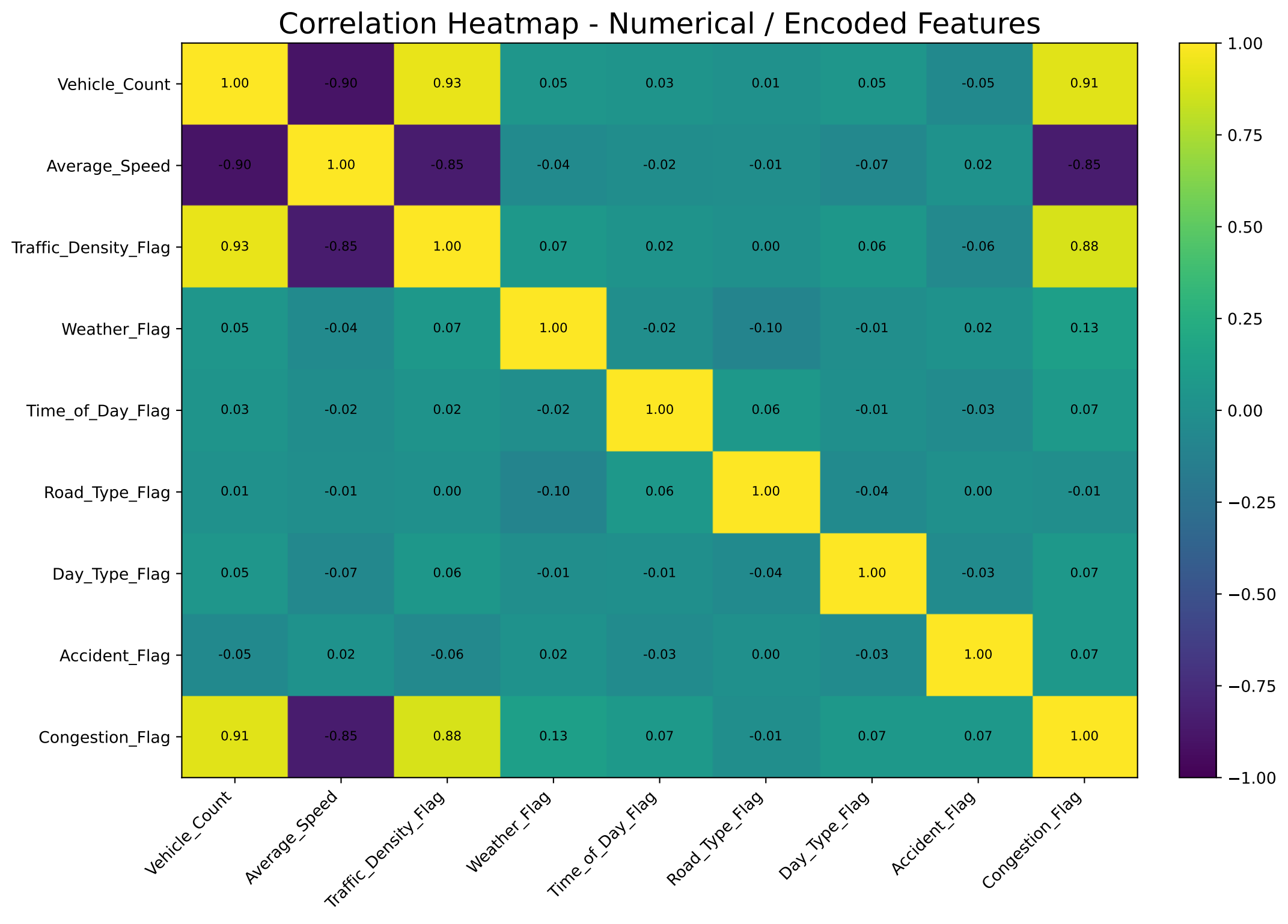
Rows: 500
Columns: 9

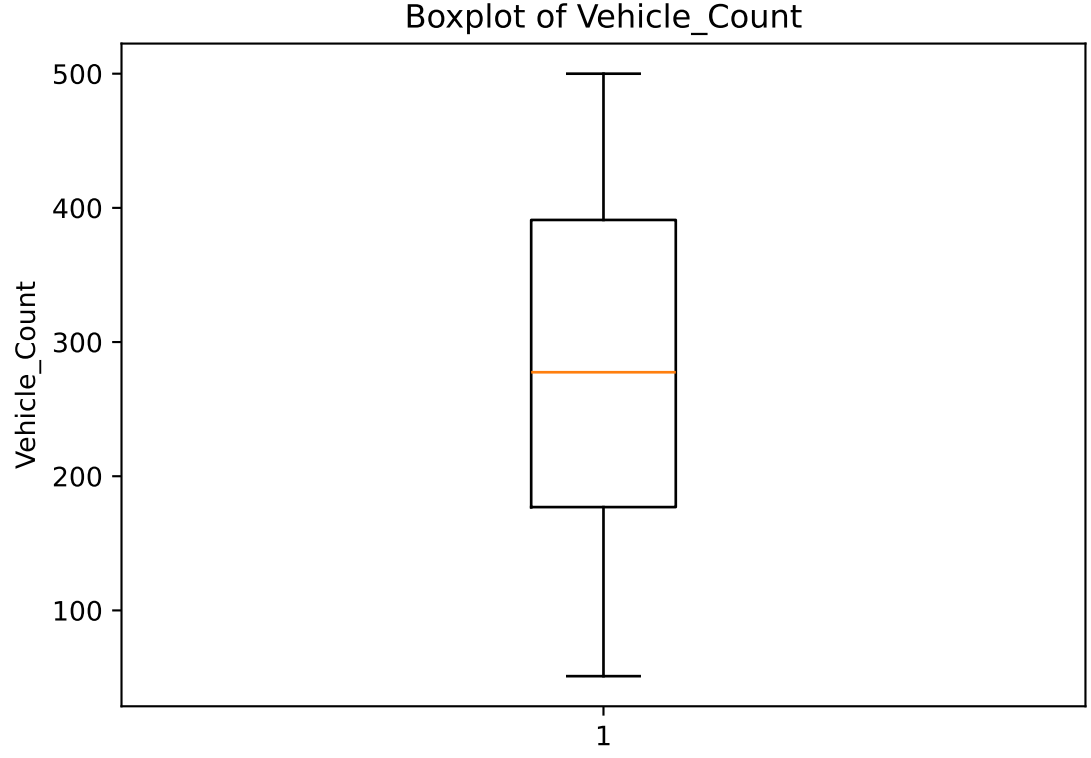
Target Variable
Congestion_Level

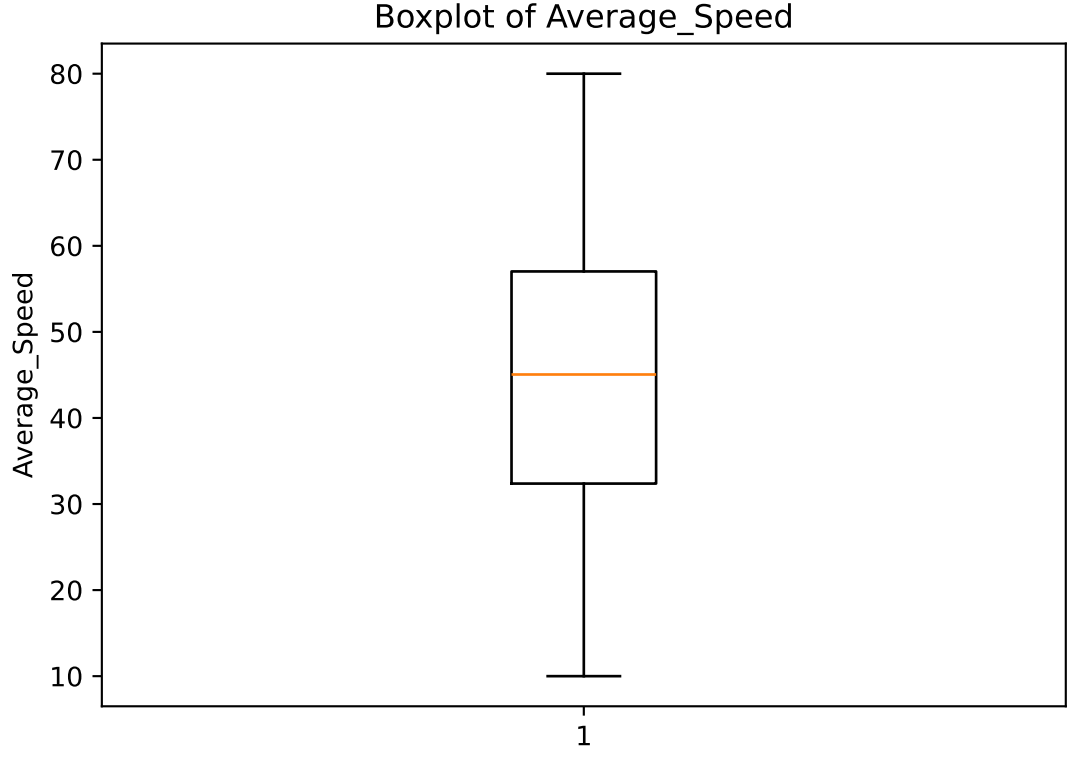
Classes
Low, Medium, High

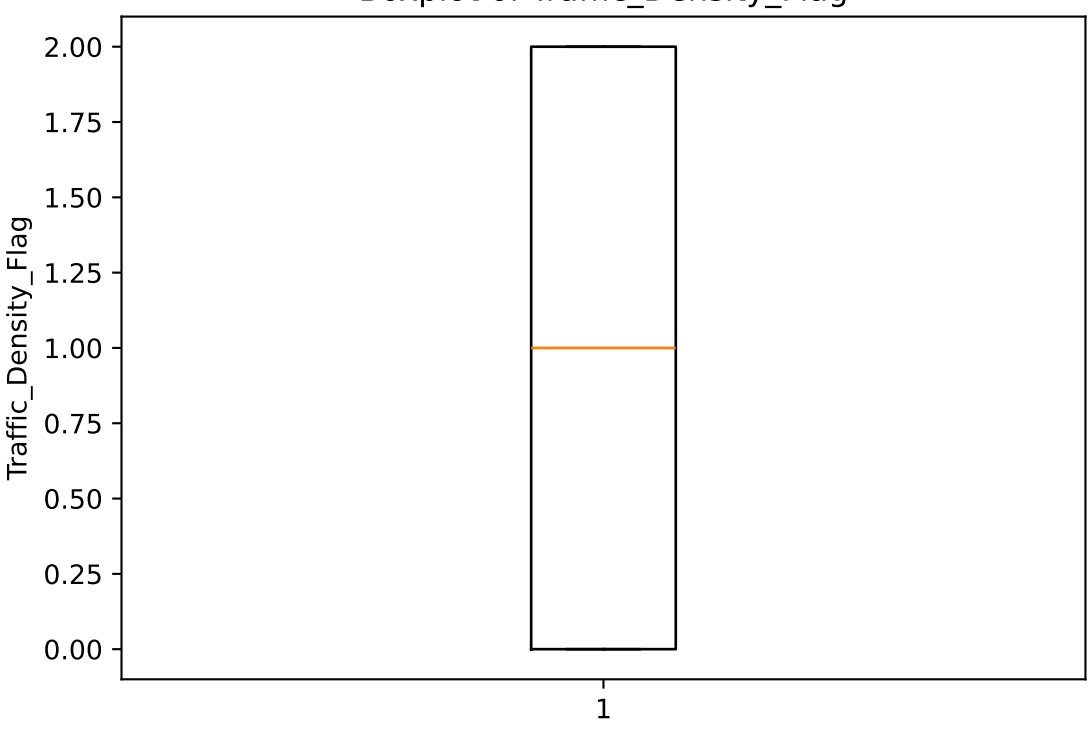

Missingness Heatmap - Traffic Congestion Dataset
Vehicle_Count
Average_Speed
Traffic_Density
Weather
Time_of_Day
Road_Type
Day_Type
Accident
Congestion_Level
Rows

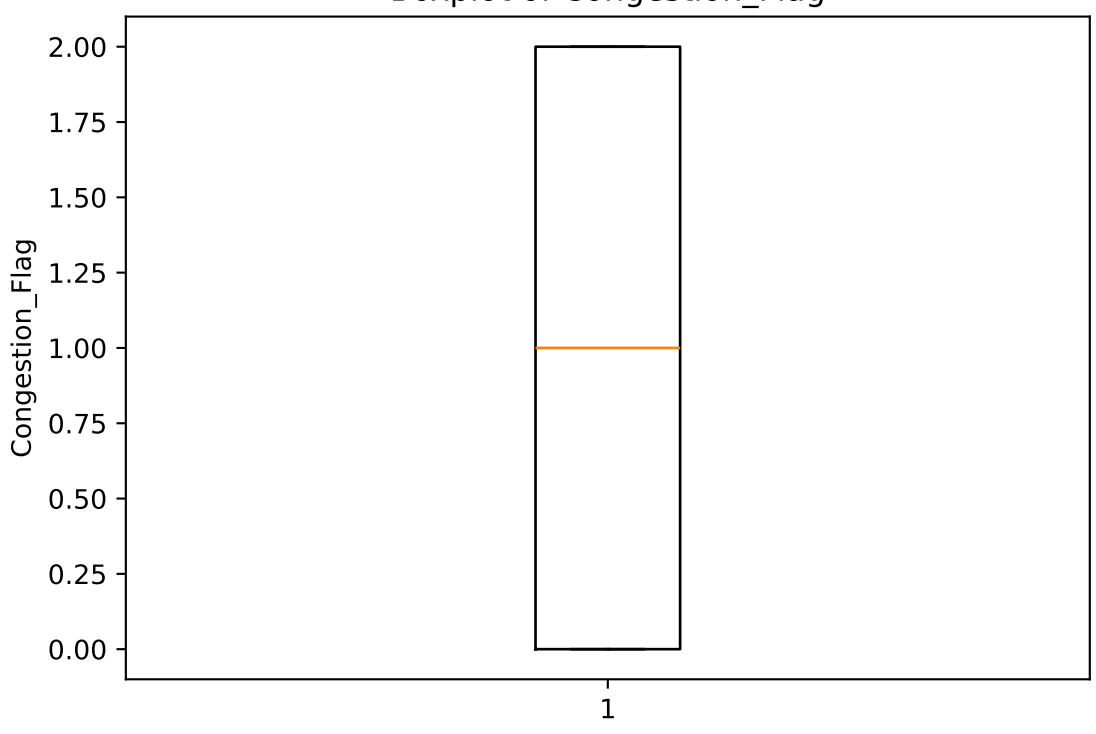

Congestion Level Distribution
Count
Congestion Level
Low
Medium
High
191
127
182
0
25
50
75
100
125
150
175
200


Correlation Heatmap - Numerical / Encoded Features
Vehicle_Count 1.00 -0.90 0.93 0.05 0.03 0.01 0.05 -0.05 0.91
Average_Speed -0.90 1.00 -0.85 -0.04 -0.02 -0.01 -0.07 0.02 -0.85
Traffic_Density_Flag 0.93 -0.85 1.00 0.07 0.02 0.00 0.06 -0.06 0.88
Weather_Flag 0.05 -0.04 0.07 1.00 -0.02 -0.10 -0.01 0.02 0.13
Time_of_Day_Flag 0.03 -0.02 0.02 -0.02 1.00 0.06 -0.01 -0.03 0.07
Road_Type_Flag 0.01 -0.01 0.00 -0.10 0.06 1.00 -0.04 0.00 -0.01
Day_Type_Flag 0.05 -0.07 0.06 -0.01 -0.01 -0.04 1.00 -0.03 0.07
Accident_Flag -0.05 0.02 -0.06 0.02 -0.03 0.00 -0.03 1.00 0.07
Congestion_Flag 0.91 -0.85 0.88 0.13 0.07 -0.01 0.07 0.07 1.00
Vehicle_Count
Average_Speed
Traffic_Density_Flag
Weather_Flag
Time_of_Day_Flag
Road_Type_Flag
Day_Type_Flag
Accident_Flag
Congestion_Flag
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00


Boxplot of Vehicle_Count
Vehicle_Count
500
400
300
200
100
1


Boxplot of Average_Speed
Average_Speed
80
70
60
50
40
30
20
10
1


Boxplot of Traffic_Density_Flag
Traffic_Density_Flag
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
1


Boxplot of Congestion_Flag
Congestion_Flag
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
1